



PowerPoint Presentation by
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MANAGEMENT ACCOUNTING

8th EDITION

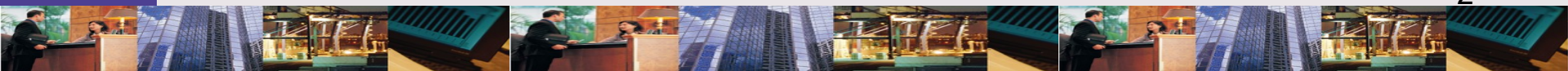
BY

HANSEN & MOWEN

17 ENVIRONMENTAL COST MANAGEMENT

LEARNING OBJECTIVES

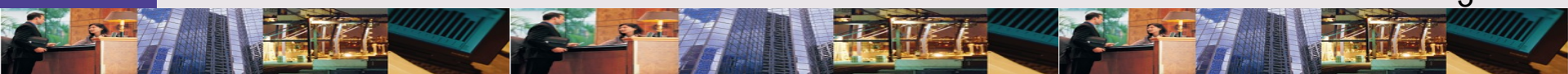
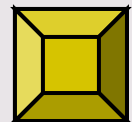
After studying this chapter, you should be able to:



LEARNING OBJECTIVES

1. Discuss the importance of measuring environmental costs.
2. Explain how environmental costs are assigned to products & processes.
3. Describe the life-cycle cost assessment model.
4. Compare & contrast activity- & strategic-based environmental control.

*Click the button to skip
Questions to Think About*



QUESTIONS TO THINK ABOUT: Thamus, Inc.

What are environmental costs?
Are environmental costs
significant enough to track &
report to management?

QUESTIONS TO THINK ABOUT: Thamus, Inc.

Will improving environmental
performance increase or
decrease total environmental
costs?



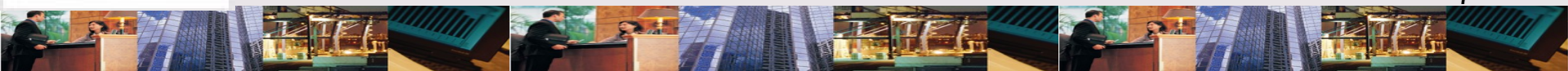
QUESTIONS TO THINK ABOUT: Thamus, Inc.

Should environmental costs be
assigned to products &
processes as a separate item?



QUESTIONS TO THINK ABOUT: Thamus, Inc.

What is the best way to control
environmental costs?



QUESTIONS TO THINK ABOUT: Thamus, Inc.

Should companies be concerned about environmental costs that they cause but for which they do not have financial responsibility?

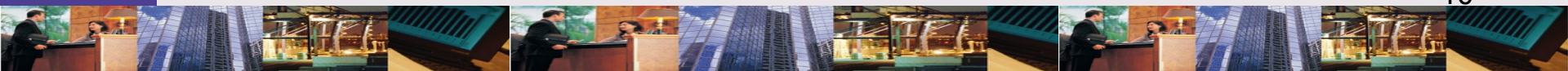
LEARNING OBJECTIVE

1

Discuss the importance of measuring environmental costs.

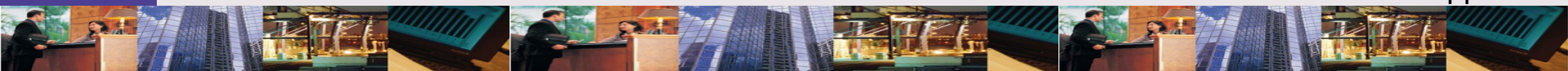
Why is it important to
measure environmental
costs?

**Awareness of environmental costs
is important because
environmental regulations &
fines have increased.**



ECOEFFICIENCY: Definition

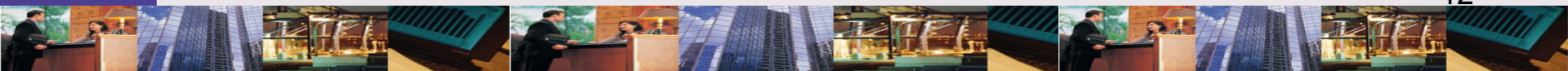
Maintains that producing more useful goods, services is consistent with reducing negative environmental impacts.



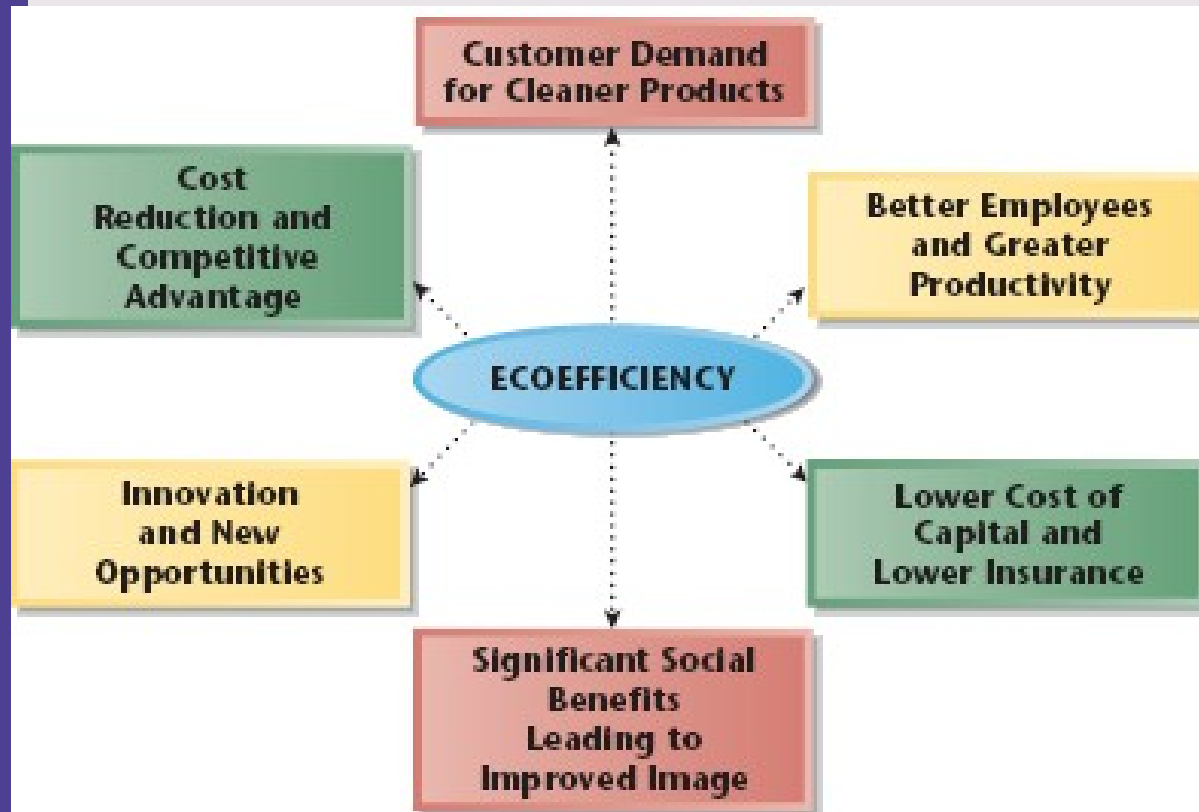
SUSTAINABLE DEVELOPMENT:

Definition

Is development that meets needs of present without compromising ability of future generations to meet their own needs.



ECOEFFICIENCY

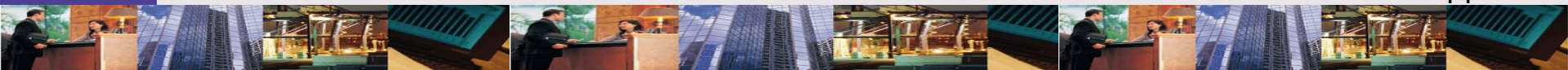


Many things provide causes & incentives that foster ecoefficiency.

EXHIBIT 17-1

ENVIRONMENTAL QUALITY COST MODEL

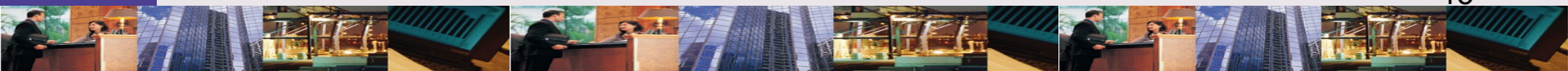
Looks at costs and their impact for damage done to the environment. In addition to direct costs, there are costs to preventing environmental degradation.



ENVIRONMENTAL DETECTION COSTS

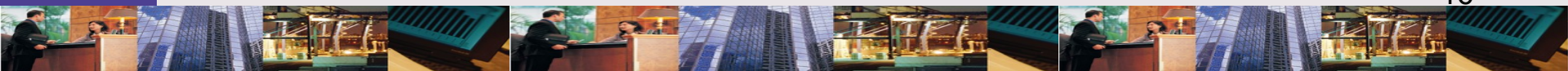
Are costs to determine compliance with appropriate environmental standards including:

- ✳️ Regulatory government laws
- ✳️ Voluntary standards (ISO 14001)
- ✳️ Management's environmental policies



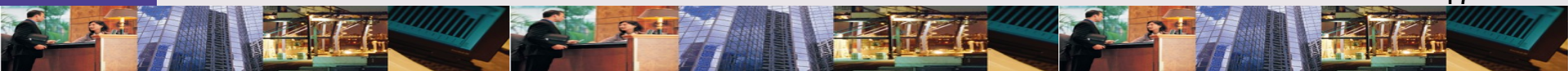
What are environmental external failure costs?

Environmental external failure costs are costs of activities performed **after discharging contaminants & waste into the environment.**



What information does an environmental cost report provide?

**Environmental cost reports reveal
1) the impact of environmental
costs on firm profitability & 2)
relative amounts expended in
each category.**



ENVIRONMENTAL COST REPORT

External failure costs are the **largest** costs.

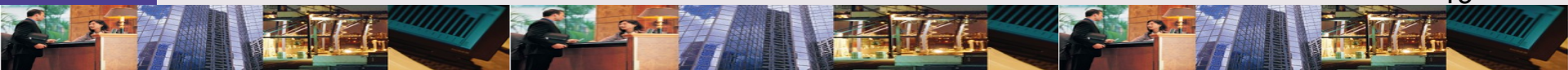
Thamus Corporation Environmental Cost Report For the Year Ended December 31, 2008			
Environmental Costs			Percentage of Operating Costs
Prevention costs			
Training employees	\$ 600,000		
Designing products	1,800,000		
Selecting equipment	<u>400,000</u>	\$ 2,800,000	1.40%
Detection costs			
Inspecting processes	\$2,400,000		
Developing measures	<u>800,000</u>	3,200,000	1.60
Internal failure costs			
Operating pollution equipment	\$4,000,000		
Maintaining pollution equipment	<u>2,000,000</u>	6,000,000	3.00
External failure costs			
Cleaning up lake	\$9,000,000		
Restoring land	5,000,000		
Incurring property damage claim	<u>4,000,000</u>	<u>18,000,000</u>	<u>9.00</u>
Totals		<u>\$30,000,000</u>	<u>15.00%</u>

Exhibit 17-3 Environmental Cost Report

EXHIBIT 17-3

Can environmental failure costs be reduced?

Yes! Investing more in prevention & detection activities will reduce environmental failure costs.



ENVIRONMENTAL FINANCIAL STATEMENT

Thamus Corporation Environmental Financial Statement For the Year Ended December 31, 2008

Environmental benefits	
Cost reductions, contaminants	\$ 3,000,000
Cost reductions, hazardous waste disposal	4,000,000
Recycling income	2,000,000
Energy conservation cost savings	1,000,000
Packaging cost reductions	<u>1,500,000</u>
→ Total environmental benefits	<u>\$ 11,500,000</u>
Environmental costs	
Prevention costs	\$ 2,800,000
Detection costs	3,200,000
Internal failure costs	6,000,000
External failure costs	<u>18,000,000</u>
→ Total environmental costs	<u>\$30,000,000</u>

Investments in environmental benefits partially offset environmental costs.

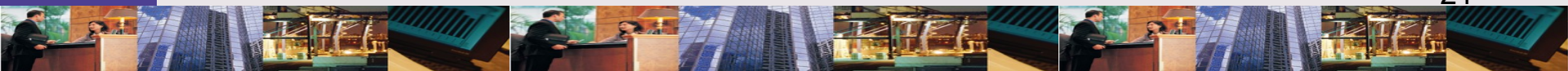
Exhibit 17-4 Environmental Financial Statement

EXHIBIT 17-4

LEARNING OBJECTIVE

2

Explain how
environmental costs
are assigned to
products & processes.



ASSIGNING ENVIRONMENTAL COSTS

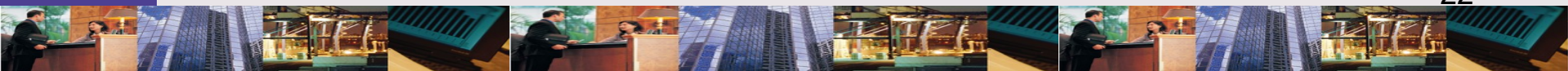
✧ Product costs

- ✧ Packaging

- ✧ Products themselves

✧ Process costs

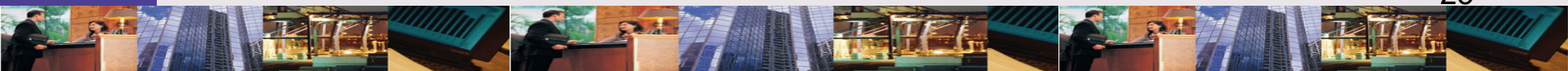
- ✧ Solid, liquid, gaseous residues



ENVIRONMENTAL COSTING:

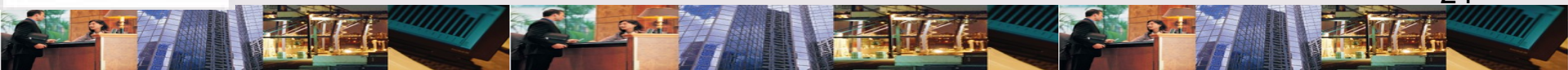
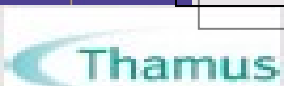
Definition

Full environmental costing assigns both private & societal costs to products. Private costs are caused by internal processes.



THAMUS, INC.: Background

Two approaches can be used to assign environmental costs to products: functional-based or activity-based costing. Environmental costs, often hidden in overhead are separated out to assign to products. Functional-based costing may work well for homogeneous products, but Thamus is a diversified, multi-product firm.



FORMULA: Cadmium Example

Costs are assigned proportionately when multiple products are produced.

External failure cost:

= Total failure cost ÷ Units produced

= \$150,000 / 20,000 = \$7.50 per unit



LEARNING OBJECTIVE

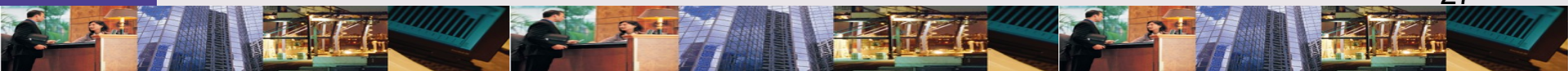
3

Describe the life-cycle
cost assessment model.

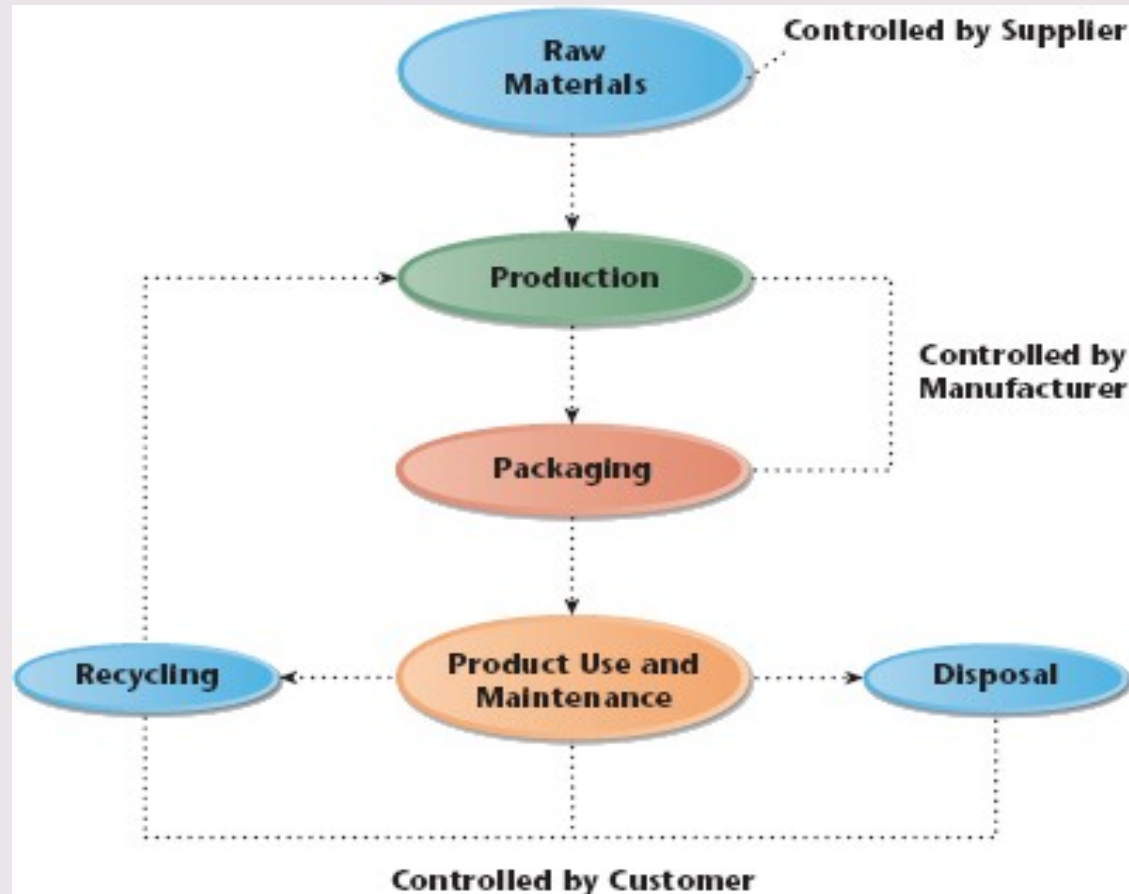
LIFE-CYCLE ASSESSMENT:

Definition

Identifies environmental consequences of a product through its entire life cycle & searches for improvements.



PRODUCT LIFE-CYCLE STAGES



Some costs are controlled by the supplier while others are controlled by customers.

EXHIBIT 17-6

ASSESSMENT STAGES

✧ 3 formal stages

✧ Inventory analysis

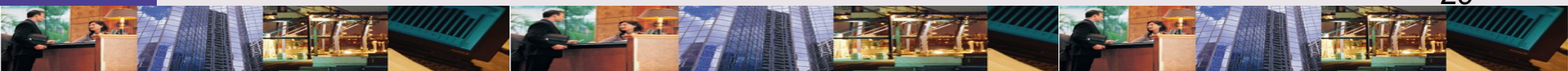
- ✧ Types, quantities inputs needed
- ✧ Environmental releases

✧ Impact analysis

- ✧ Effects of competing designs
- ✧ Relative ranking of effects

✧ Improvement analysis

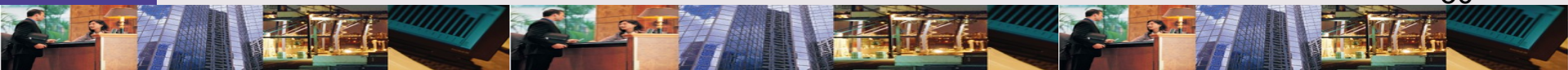
- ✧ Objective: to reduce environmental impacts



LEARNING OBJECTIVE

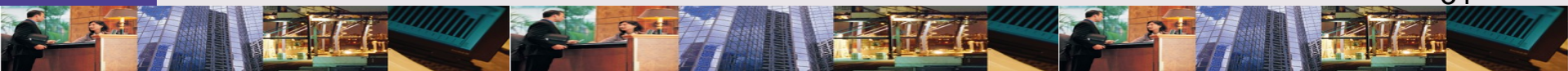
4

Compare & contrast activity- & strategic-based environmental control.



ENVIRONMENTAL PERSPECTIVE

- ✧ 5 objectives for environmental perspective
 - ✧ Minimize use of raw or virgin materials
 - ✧ Minimize use of hazardous materials
 - ✧ Minimize energy requirements for production, use of product
 - ✧ Minimize release of solid, liquid, gaseous residues
 - ✧ Maximize opportunities to recycle



OBJECTIVES & PERSPECTIVES

Companies need measures to evaluate whether objectives of the environmental perspective are being met.

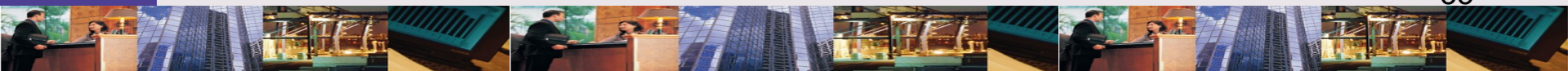
Objectives	Measures
Minimize hazardous materials	Types and quantities (total and per unit) Percentage of total materials cost Productivity measures (output/input)
Minimize raw or virgin materials	Types and quantities (total and per unit) Productivity measures (output/input)
Minimize energy requirements	Types and quantities (total and per unit) Productivity measures (output/input)
Minimize release of residues	Pounds of toxic waste produced Cubic meters of effluents Tons of greenhouse gases produced Percentage reduction of packaging materials
Maximize opportunities to recycle	Pounds of materials recycled Number of different components Percentage of units remanufactured Energy produced from incineration

Exhibit 17-8 Objectives and Measures: Environmental Perspective

EXHIBIT 17-8

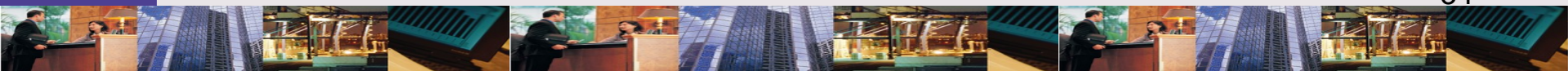
Are environmental activities
non-value-added?

**Because environmental pollution
is equivalent to economic
inefficiency, all failure activities
are non-value-added.**

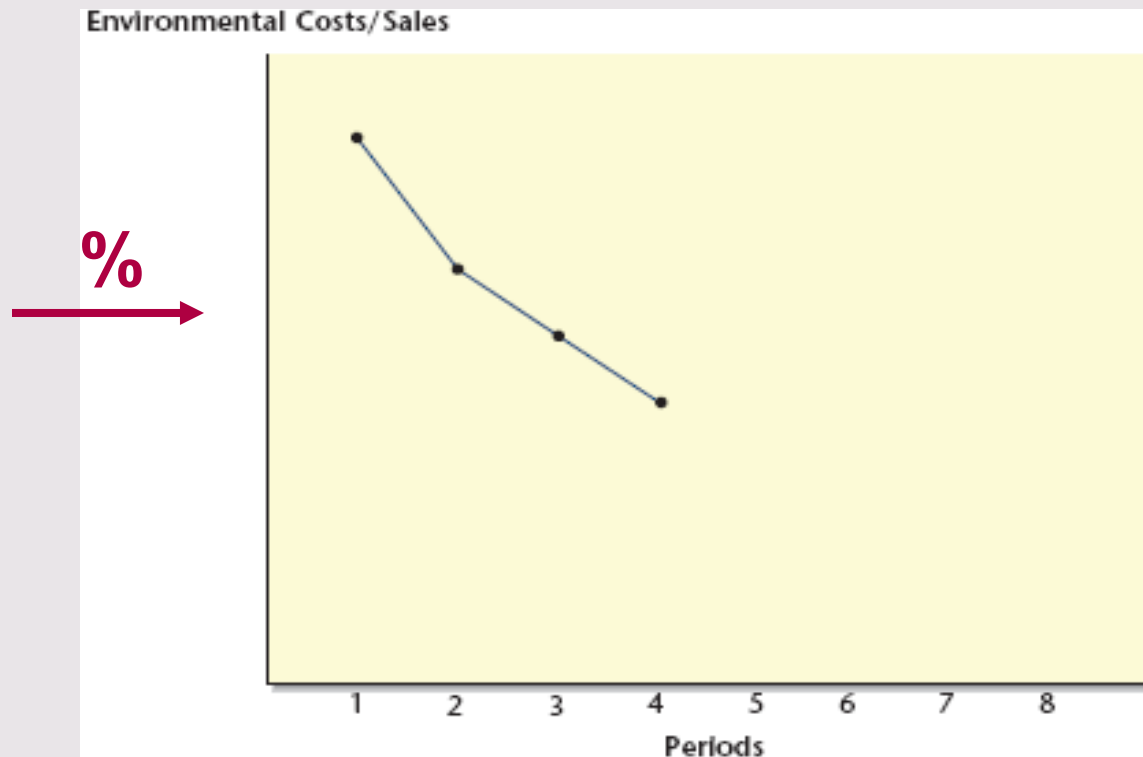


What are some good ways to
examine effects of
environmental activities?

**Graphical presentations are
effective in depicting positive
benefits of environmental.**



ENVIRONMENTAL COST TREND GRAPH



Costs as a percentage of sales trend downward over time.

EXHIBIT 17-10

BAR GRAPH FOR TRENDS

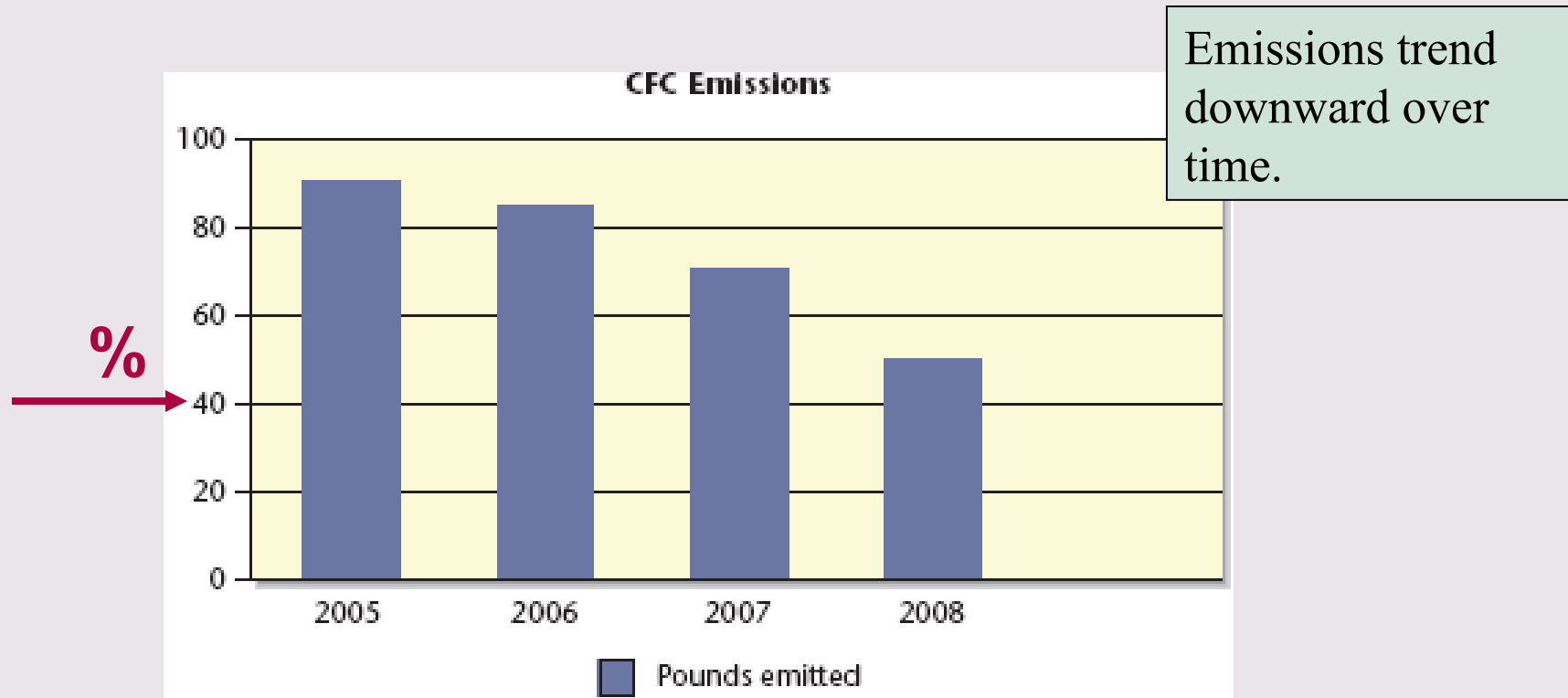
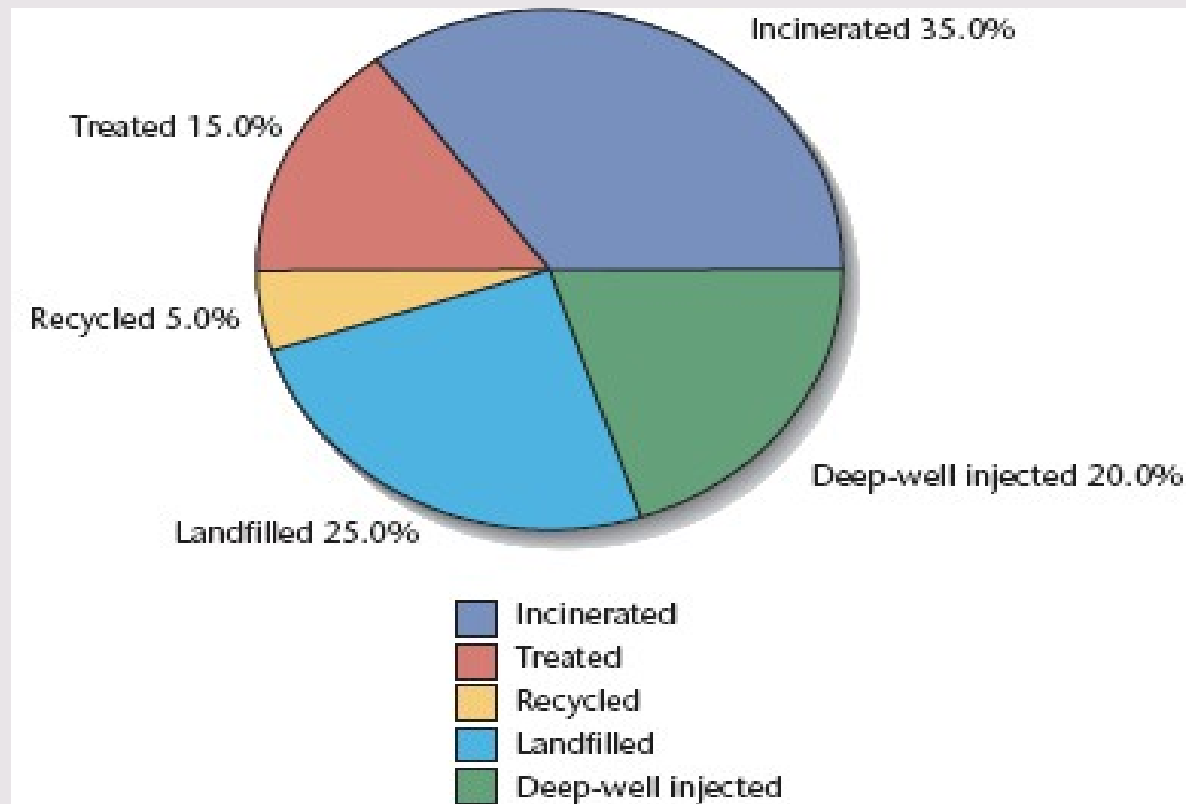


EXHIBIT 17-11

HAZARDOUS WASTE PIE CHART



Pie chart depicts proportional hazardous wastes.

EXHIBIT 17-12

CHAPTER 17

THE END